



Performance of SiO₂-CaO-B₂O₃-Al₂O₃ glass-ceramic seal under vapor atmosphere for solid oxide electrolysis cell

Xiaochun Wang ^{a c}  , Yaocheng Wang ^a, Shen Jiang ^a, Siyuan Dang ^b, Li Li ^a, Jian Pu ^b, Jiangtao Feng ^c

Show more 

 Outline |  Share  Cite

<https://doi.org/10.1016/j.ceramint.2025.09.255> 

[Get rights and content](#) 

Full text access

Abstract

The performance of Solid Oxide Electrolysis Cells (SOECs) is closely related to the gas tightness, electrical insulation, and stability of glass ceramic seals. This study shows that adding 20wt% Al₂O₃ to SiO₂-CaO-B₂O₃-Al₂O₃ glass forms the H3-20 seal, which significantly enhances these properties. The H3-20 seal has a leakage rate below 0.00043 sccm cm⁻¹ in a 90vol% vapor atmosphere at 20.7kPa and remains at 0.0007 sccm cm⁻¹ between 650 and 800°C. Its resistivity exceeds 10⁴Ωcm from 300°C to 800°C and stabilizes at 1.4×10⁷Ωcm under 1.7V and 70vol% vapor. The seal also demonstrates excellent chemical compatibility and thermal stability, maintaining a leakage rate of 0.00065 sccm cm⁻¹ after 20 thermal cycles at 750°C. These improvements are attributed to CaSiO₃ crystallization and a silicic acid gel film that fills gaps and enhances sealing performance.



Keywords

Solid oxide electrolysis cell; Glass ceramic seal; Leakage rate; Insulation; Sealing mechanism

1. Introduction

Solid oxide cells (SOCs) have attracted much attention in research because of high energy conversion efficiency and wide fuel selectivity [1]. Solid oxide electrolysis cell (SOEC) as electrochemical device that is able to electrolyzing water into hydrogen and oxygen which occurs at the triple-phase boundary of the fuel electrode [2]. Seal as the key component to exhibits excellent resistance to high temperature water vapor corrosion, it should have good gas tightness, thermal mechanical property and long stability. Glass and glass ceramic seals are promising candidates for SOEC stack because of several crucial characteristics that are required for seals applications, such as (1) variable glass transition temperature is compatible with the operating range of an SOFC, i.e., 650–800 °C; (2) coefficient of thermal expansion (CTE) similar to that of other fuel cell components; (3) suitable chemical resistance against the formation of crystalline phase, which can promote the rupture of the device during its operation.

The vapor corrosion will damage the glass structure, characteristic temperature and microstructure resulting in interface chemical compatibility, long term sealing performance and high-temperature insulation between the seals and adjacent at high temperature [3], [4], [5], [6]]. M.K. Mahapatra et al. found that the B₂O₃:SiO₂ ratio increases from 0wt% to 1.4wt%, the T_g of SrO–La₂O₃–Al₂O₃–B₂O₃–SiO₂ glasses decreased from 775 °C to 635 °C, and the T_d decreased from 815 °C to 670 °C as well, further more, glass crystallization is accelerated when they are exposed to high temperature steam, compared with the steam-free atmosphere [7], [8], [9]]. Blaszcak et al., investigated the impact of incorporating a minor component of cobalt on the conversion efficiency of co-electrolysis of H₂O/CO₂ in the SOEC mode, and discovered that the addition of 3.6wt% Co to the Ni-YSZ cermet material of the cells via the wet impregnation method, methane concentration exceeding 2.5 times that of samples without cobalt. thus enhancing the co-electrolysis of H₂O and CO₂ [10]. Wang et al. investigated the thermal stability of SrO–La₂O₃–Al₂O₃–SiO₂ glass ceramic seals under the input pressure condition of 20.7kPa at the temperature range of 850–900 °C that when glass samples are

exposed to high-temperature vapors or when the temperature rises to the point when the inter-connection between [SiO₄] tetrahedral units are disrupted, surface crystallization occurs, thereby increasing the microstructural heterogeneity within the glass network [11]. A further study by Sabato et al. designed that the suitable resistivity value for the Na-containing diopside-based glass ceramic seal is slightly lower than 10⁶Ωcm and is about in the range of 10⁵–10⁶Ωcm while it subjected to a voltage of 1.3V under a oxidizing on one side and reducing on the other atmosphere environment [12]. It can be known that although above researchers have conducted various studies on different seals, there are few studies on the performance of glass ceramic seal in vapor atmosphere. This is because the complexity of such environments, which pose numerous technical challenges and research difficulties. It is necessary to conduct a research on their characteristics and evaluate the applicability of glass ceramic seal in SOEC.

The advantages of SiO₂-CaO-B₂O₃-Al₂O₃ glass-ceramic seal mainly include excellent mechanical properties, thermal stability, sealing performance and relatively small crystalline characteristics. First of all, SiO₂-CaO-B₂O₃-Al₂O₃ glass-ceramic seal can prevent seal deformation because of the high hardness and strength of Al₂O₃, and thus Al₂O₃ act as a reinforcing agent to improve the mechanical stress between the SiO₂-CaO-B₂O₃-Al₂O₃ glass-ceramic seal and adjacent components [13]. Secondly, the glass ceramic seal has excellent high temperature stability in the long term operate environments without degradation due to the addition of Al₂O₃ ceramic material [14]. In addition, the addition of 20wt% Al₂O₃ ceramic materials to SiO₂-CaO-B₂O₃-Al₂O₃ glass system will reduces the porosity and enhance the gas tightness of glass ceramic seal. Due to their similar thermal expansion coefficients, the seal can maintain excellent structural stability and sealing property [13]. Finally, because the moderate Al₂O₃ ceramic is embedded in SiO₂-CaO-B₂O₃-Al₂O₃ glass system acting as frame to enhance structural stability, the SiO₂-CaO-B₂O₃-Al₂O₃ glass phase own proper fluidity, which can infiltrate components and ceramic without wide flow and release thermal stress during sealing process [15]. The results of the preliminary experiments indicate that H3 glass and Al₂O₃ have good wettability at high temperature. Moreover, the addition of 20wt% Al₂O₃ to H3 glass can significantly improve the mechanical and sealing properties. However, although SiO₂-CaO-B₂O₃-Al₂O₃ glass ceramic seal (H3-20) has been considered as sealing material in SOFCs due to its excellent performances as mentioned previously [16], glass microstructure and properties, such as gas tightness, interface compatibility, chemical stability, insulation, and long-term stability, will changed under high vapor environment. In this study, we aim to investigate the effects of temperature, input gas pressure and water vapor ratio on the air tightness, microstructure, chemical compatibility, insulation, long term stability and sealing mechanism of H3-20 glass ceramic seal in vapor atmosphere.

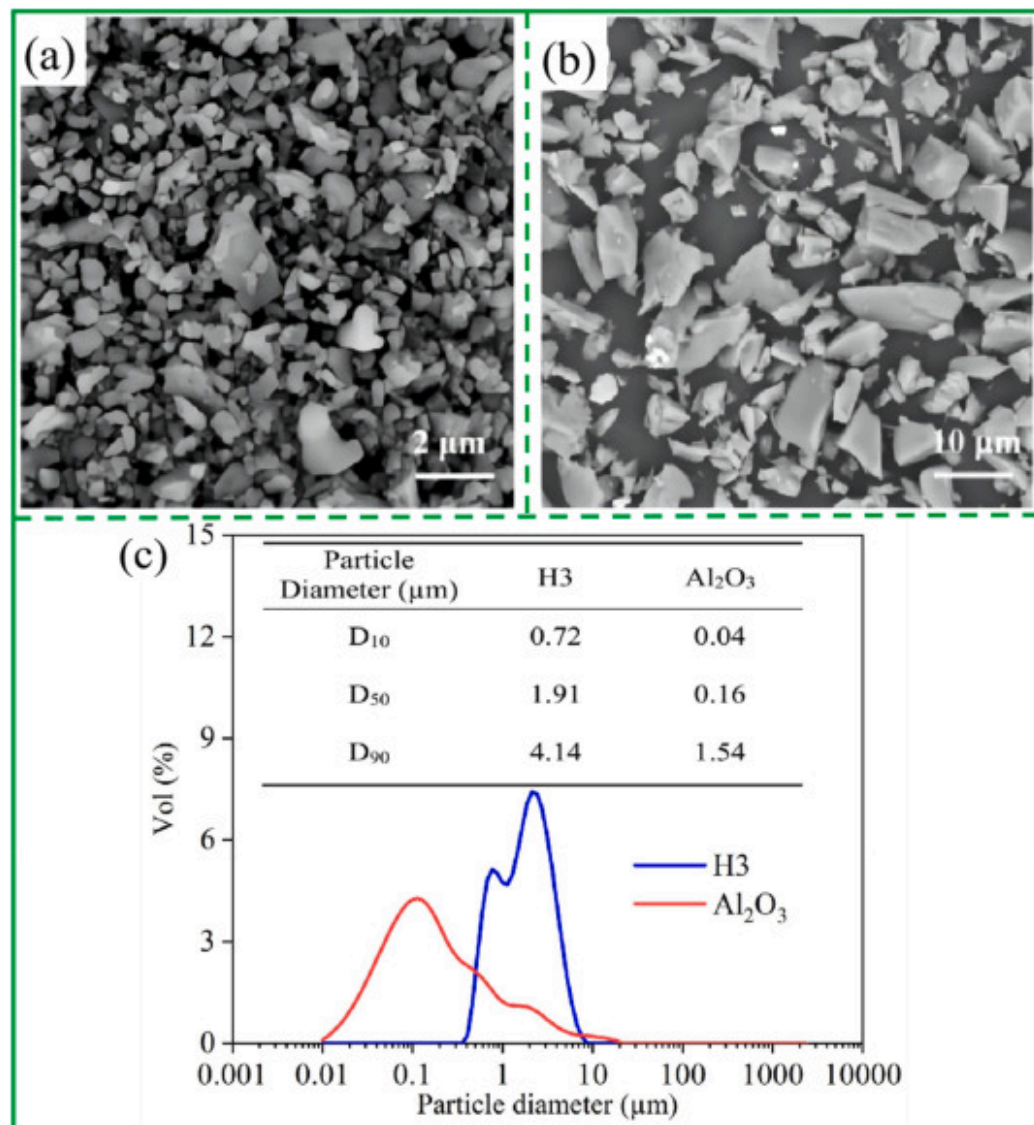
2. Experimental

2.1. Seal preparation

The glass ceramic seals were prepared by adding 20wt% Al₂O₃ into CaO-Al₂O₃-B₂O₃-SiO₂ system glass (H3) which was designed as H3-20. H3 glass powdered glasses were obtained from a commercial supplier (Institute of Process Engineering, Chinese Academy of Sciences). The glass and ceramic powder mixtures were ball-milled for 6h in ethyl alcohol to obtain a uniform mixture, and then adding some organic additions, including solvent, dispersant, polyvinyl butyral (PVB) and phthalate and so on. After ball milling around 24h, the slurry was tape casted to obtain a seal strip with thickness of around 0.4mm after drying at room temperature. The preparation process of H3-glass ceramic seal tape is shown in [Fig. 2\(a and b\)](#).

2.2. Testing and characterization

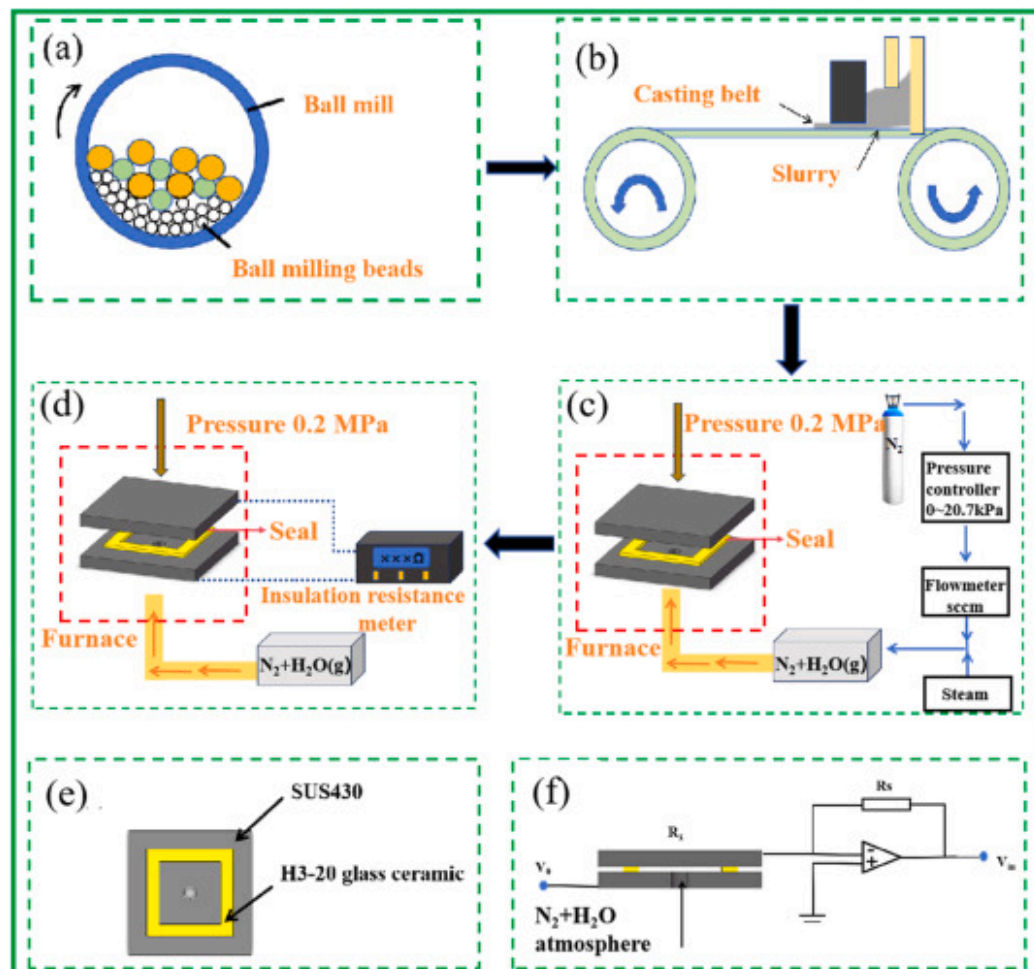
The particle size of H3 glass powders and Al₂O₃ particles was measured by Mastersize-v3.50 laser particle size analyzer. [Fig. 1](#) shows the micrographs and particle size distribution of H3 glass and Al₂O₃. It is clear can be seen from [Fig. 1\(a, b\)](#) that average particles size of H3 glass powders are about 2μm and the Al₂O₃ powders are about 0.1μm, respectively as given in corresponding micrographs in [Fig. 1\(c\)](#).



[Download: Download high-res image \(807KB\)](#)

[Download: Download full-size image](#)

Fig. 1. Microscopic morphology and particle size distribution of the powder samples (a) SEM image of Al₂O₃, (b) SEM image of H3, (c) particle size distribution of Al₂O₃ and H3.



[Download: Download high-res image \(682KB\)](#)

[Download: Download full-size image](#)

Fig. 2. The schematic diagram of H3-20 compressive composite seal preparation (a,b), leakage rate test (c), resistance insulation test in dual atmosphere (d), illustrative schematic of resistance testing procedure (e) and schematic of sample (f).

The seals were cut into a square window frame with an outside dimension of 70×70mm and inside dimension of 50×50mm for leakage rates test. The H3-20 seal was placed between two of SUS430 alloy plate, as shown in Fig. 2(c). A digital flow meter (ALICAT 66735) with an accuracy of 0.0001 sccm was chosen to measure the leakage rates with mixture of N₂ and water vapor as

testing gas. The details and principles of the test equipment have been described by previous researchers of our group in Ref. [17]. To investigate the leakage reasons, the surface of the tested seal samples was observed by a scanning electron microscope (SEM) to explore the erosion effects of steam.

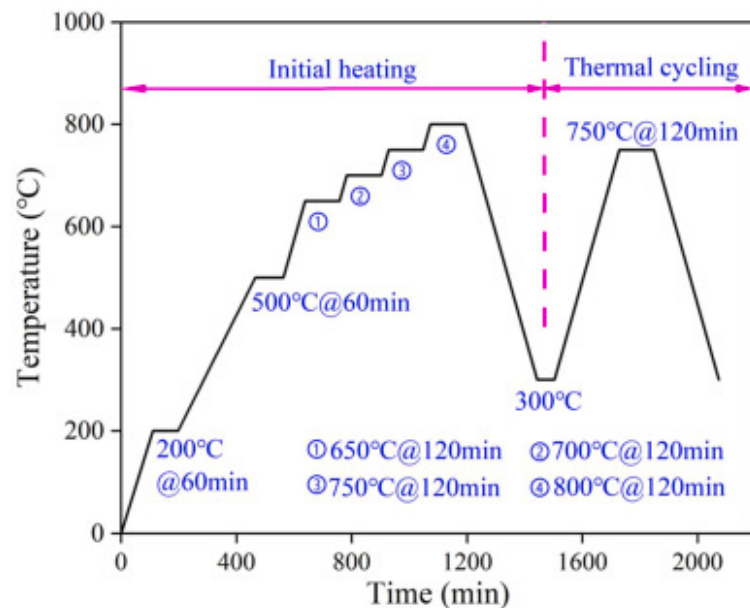
The influence of steam on the resistivity of H3-20 seals was measured by JK283 insulation resistance meter, as shown in Fig. 2(d, e) The SUS430/seal/SUS430 joints were heated from room temperature to 800°C in a 70vol% steam atmosphere with an applied voltage of 1.7V under 0.2MPa and the resistance at different temperature segments was obtained by connecting the point welded nickel-chromium wires to the resistance meter. H3-20 glass ceramic seal was placed between interconnect (SUS430 alloy) and cell anode (NiO-YSZ) under a compressive load of 0.2MPa and 70vol% water vapor at 750°C for 2h. Energy dispersion spectrometer (EDS) was used to analyze the chemical composition of the seals.

The insulation resistance test was conducted using a negative feedback amplification circuit formed by the tested resistance R_x and the feedback resistance R_s of the operational amplifier, as shown in Fig. 2(f). According to Kirchhoff's current law [18], the circuit was solved to obtain the tested resistance, and the calculation formula is given by formula (1)

$$R_x = \frac{-V_0 R_s}{V_{in}} \quad (1)$$

where V_0 is the input voltage, R_x is the tested resistance, R_s is the precision resistance, and V_{in} is the output sampling voltage.

Fig. 3 shows the heating process of H3-20 glass ceramic seal. The seals were heated at a rate of 2°C min⁻¹ from room temperature to 200°C and held for 60min to ensure a uniform temperature environment. Then, the temperature was increased at a rate of 1°C min⁻¹ to 500°C and held for 60min to allow the organic substances to volatilize completely, avoiding the formation of pores due to the rapid volatilization of residual organic substances, which could increase the leak rate. Finally, the temperature was increased at a rate of 2°C min⁻¹ to 650°C, 700°C, 750°C, and 800°C, and held for 120min to test the leakage rate under different input gas pressures.



[Download: Download high-res image \(323KB\)](#)

[Download: Download full-size image](#)

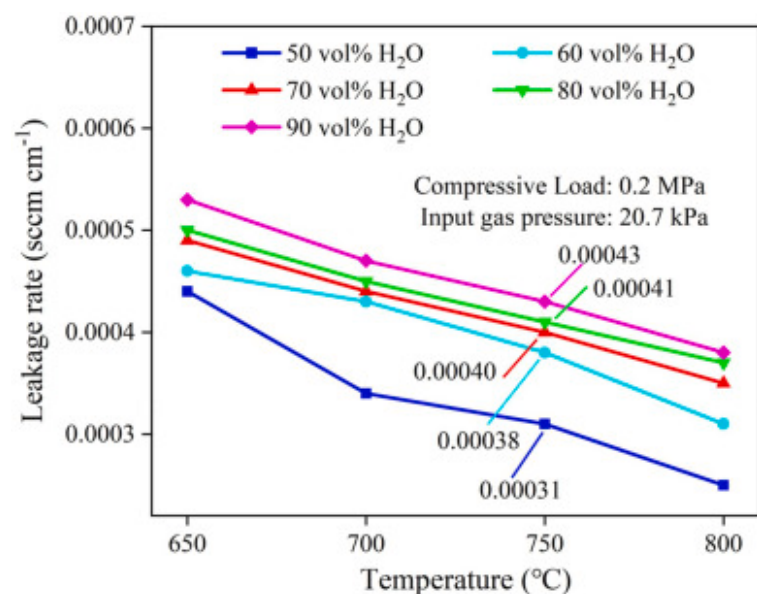
Fig. 3. Temperature profile for initial heat-up, holding treatment and subsequent thermal cycles.

3. Results and discussion

3.1. The effect of vapor content on the leakage rate of glass-ceramic seal

In order to clarify the effect of H₂O vapor addition on the leakage rate and microstructure of the H3-20 composite seals, the leakage rate, XRD and SEM of H3-20 seals are tested under the compressive load of 0.2MPa at 650°C–800°C. In addition, in practical applications of SOEC, the proportion of water vapor typically varies between 70vol% and 90vol% to ensure efficient reactions and minimize side reactions [19]. If the proportion of water vapor is too low, the maximum hydrogen production capacity of the SOEC can't be measured. Therefore, in this paper, a volume ratio range of 50vol%–90vol% for water vapor content is selected to simulate the actual operating environment of the SOEC and test the leakage rate of H3-20 glass ceramic seal. As shown in Fig. 4, the leakage rate of the H3-20 seal with different vapor ratio under a compressive pressure load of 0.2MPa and

input gas pressure of 20.7 kPa between 650°C and 800°C. The leakage rates of the H3-20 seal is gradually decreases with the temperature rises from 650°C to 800°C but shows an increasing trend with increasing vapor ratio. However, the overall leakage rate of H3-20 glass ceramic seal is below 0.0006 sccm cm⁻¹ while the vapor ratio of 90 vol%, which means that the H3-20 glass ceramic seal has excellent gas tightness. In addition, the maximum leakage rate are 0.00049 sccm cm⁻¹, 0.00044 sccm cm⁻¹, 0.00040 sccm cm⁻¹, 0.00035 sccm cm⁻¹ at 650°C, 700°C, 750°C, and 800°C, respectively while the vapor ratio of 70 vol%. It can be attributed to the increased softening and fluidity of the glass at higher temperature, which fills the pores created by the evaporation of organic additives, and the H3-20 glass ceramic seal prevent the interfacial corrosion of H₂O molecules on the seal as well. Moreover, the glass flow can fill the gaps at the interface between the interconnect and the seal, further reducing the leakage rate.



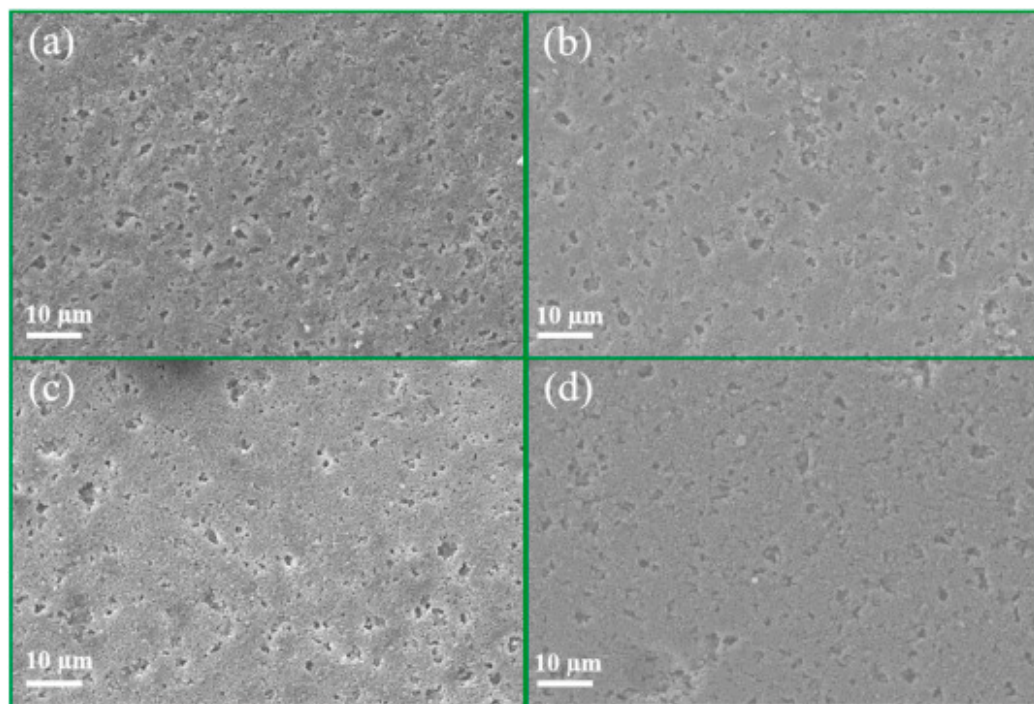
[Download: Download high-res image \(369KB\)](#)

[Download: Download full-size image](#)

Fig. 4. The leakage rate of the H3-20 seal with different vapor ratio under compressive pressure load of 0.2 MPa and input gas pressure of 20.7 kPa between 650°C and 800°C.

Fig. 5 shows the surface morphology of H3-20 seal heated at 650°C, 700°C, 750°C and 800°C in 70 vol% water vapor for 2 h. It is observed on the seals that the pores have gradually decreased with increase of temperature from 650°C to 800°C, which means

the H3-20 glass ceramic seal become denser with increasing temperature. When H3-20 glass ceramic seal lower than 750°C, pores can hardly be seen on the surface of the seal, suggesting that leakage rate are related with density of the seals. When the temperature reaches 750°C, the number and size of pores on the surface of the H3-20 glass ceramic seal gradually decrease and the edges of the are smoother. When the temperature high than 800°C, most of the pores have healed. The results show that an increase in temperature can effectively reduce the number of pores, enhance the gas tightness of the H3-20 glass ceramic seal, which has also been found on other glass based seals system [20]. The H3-20 seal exhibits high temperature structure strength, of which the glass ceramic network can effectively prevent corrosion caused by 70vol% water vapor under relatively high compressive load and temperature gradient.

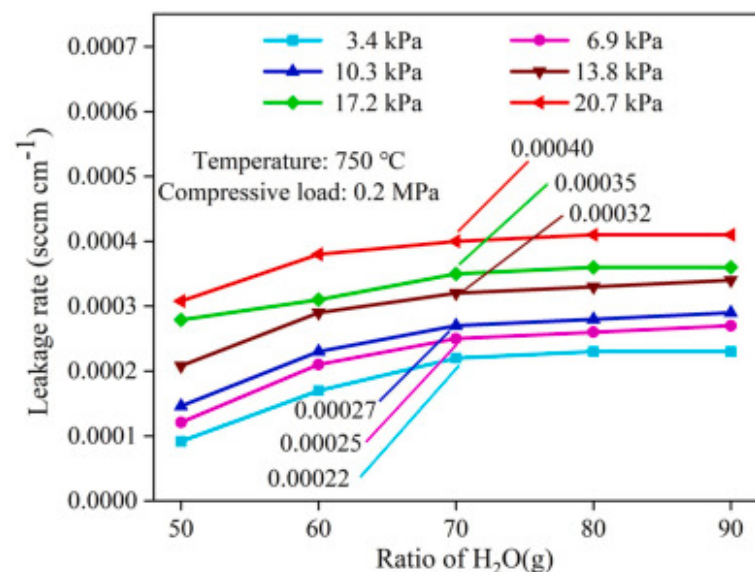


[Download: Download high-res image \(918KB\)](#)

[Download: Download full-size image](#)

Fig. 5. The SEM images of the H3-20 seal at different temperatures in 70vol% water vapor for 2 h: (a) 650°C, (b) 700°C, (c) 750°C, and (d) 800°C.

Fig. 6 shows the leakage rate of H3-20 glass ceramic seal under different vapor ratio from 50vol% to 90vol% with input gas pressure of 20.7 kPa and a compressive load of 0.2 MPa at 750 °C. It is obviously observed that the leakage rate of the seals are gradually increased with the ratio of vapor increasing from 50vol% to 70vol% and then remains stable when the ratio of vapor increasing to 90vol%. For example, when the input gas pressure is fixed at 20.7 kPa, the measured leakage rate of 50vol% vapor, 60vol% vapor, 70vol% vapor, 80vol% vapor, 90vol% vapor are 0.00031 sccm cm⁻¹, 0.00038 sccm cm⁻¹, 0.00040 sccm cm⁻¹, 0.00041 sccm cm⁻¹ and 0.00041 sccm cm⁻¹, respectively. The similar leakage rates tendency is also observed in other ratio of vapor conditions, and the maximum leakage rate of H3-20 glass ceramic seal is only 0.0004 sccm cm⁻¹ while the ratio of vapor is 90vol%, still lower than the conventional requirement of 0.04 sccm cm⁻¹ for a seal [21]. It can be explained that a certain amount of vapor can degrade the gas tightness of the H3-20 seal, yet excessive vapor has no significant effect on the leakage rate of H3-20 seal. There are two reasons for this phenomenon, on the one hand, due to excess vapor will form a layer of silicic acid gel film on the glass surface, which has adsorptive capacity to slow down the glass corrosion, reduce leakage channels and improve gas tightness [[22], [23], [24], [25]]. On the other hand, due to ion exchange occurred with the increase in vapor concentration, forming a thin film on the glass surface that hinders further corrosion [26].



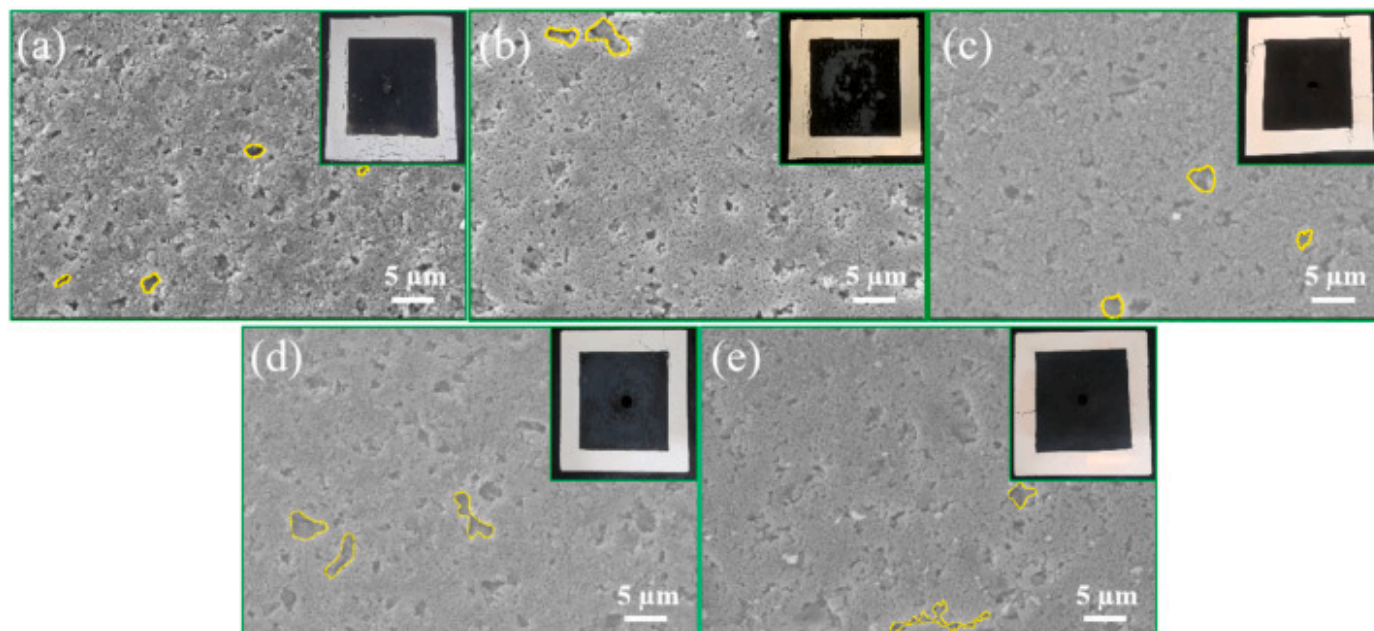
[Download: Download high-res image \(387KB\)](#)

[Download: Download full-size image](#)

Fig. 6. Leakage rate of H3-20 glass ceramic seal under different ratio of vapor from 50vol% to 90vol% at 750°C with input gas pressure from 3.4kPa to 20.7kPa under a compressive load of 0.2MPa.

Considering the electrolysis efficiency and H₂ carrying capacity of the solid oxide fuel cell, 70vol% vapor is generally used in the fuel cell stack, the relationship between the leakage rate and input pressure of H3-20 seal under 70vol% ratio of vapor is necessary to be investigated in this paper. It can be seen from Fig. 6 that when the input gas pressure from 3.4kPa to 20.7kPa under a compressive load of 0.2MPa at 750°C, the leakage rate of H3-20 glass ceramic seal are 0.00022 sccm cm⁻¹, 0.00025 sccm cm⁻¹, 0.00027 sccm cm⁻¹, 0.00032 sccm cm⁻¹, 0.00035 sccm cm⁻¹, and 0.00040 sccm cm⁻¹ while the ratio of vapor is 70vol%. The maximum leakage rate of the H3-20 glass ceramic seal is only 0.00040 sccm cm⁻¹, showing stable low leakage rates up to input gas pressure of 20.7kPa. This is because increasing input gas pressure lead to an increase in the pressure difference between the upper and lower surface of the H3-20 glass ceramic seal, causing vapor to overflow the seal and reducing its gas tightness [27].

Fig. 7 shows the macroscopic and microscopic morphology of the H3-20 glass ceramic seal in different ratio of vapor after test at 800°C. It can be seen that the number of surface pores decreased but size slightly increased as the ratio of vapor increased. Besides, the microcracks in the sealing frame gradually decrease with the increase ratio of vapor. For example, when the ratio of vapor is 50vol%, the surface exhibited dispersed and isolated small pores which is about 1.95μm, and fortunately, these small pores are not connected together to form microcracks or through-pores, as a result, the surface of the seal remains dense as before. As the ratio of vapor increased to 80vol%, the pores enlarged and became irregularly shaped due to erosion and then began to connect. Meanwhile, the size of pores is considerably increased between 3.44μm and 5.72μm. When the ratio of vapor increase to 90vol%, the surface microscopic morphology gradually smoother with interconnected pores and the pore size remained stable to that ratio of 80vol% vapor. This is due to the fact that a large amount of vapor will form a thick “silicic acid gel film” on the surface of the H3-20 glass ceramic seal, which enhances the vapor corrosion resistance that slow down its corrosion progress on the pores and other defects of the seal and inhibited leakage [28].



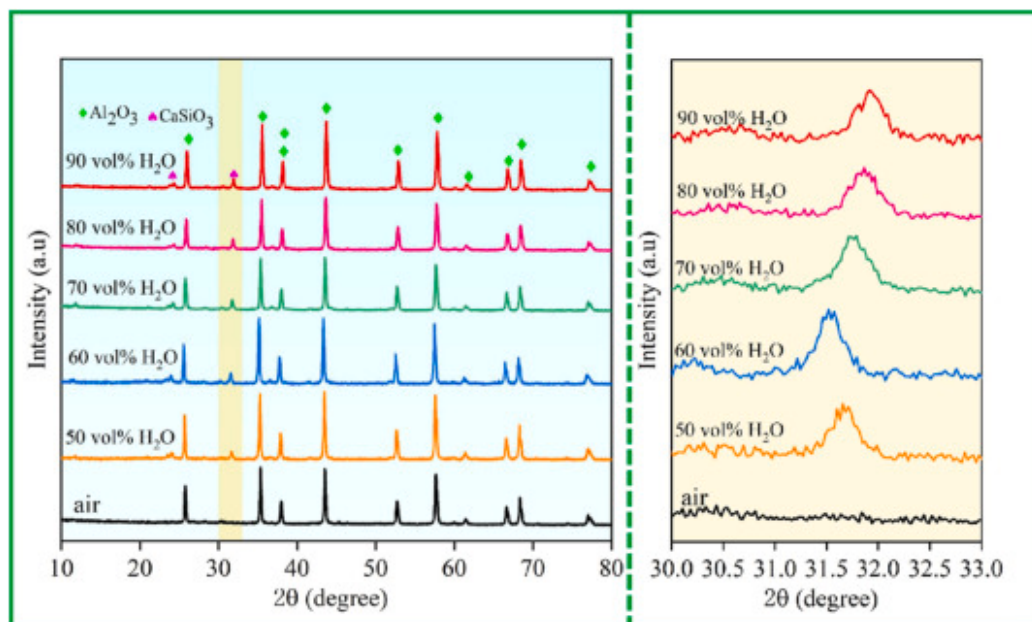
Download: [Download high-res image \(891KB\)](#)

Download: [Download full-size image](#)

Fig. 7. The macroscopic and microscopic morphology of the H3-20 glass ceramic seal in different ratio of vapor at 800°C. (a) 50vol%, (b) 60vol%, (c) 70vol%, (d) 80vol%, (e) 90vol%.

Fig. 8 shows the X-ray diffraction analysis of the H3-20 glass ceramic seal after heat treated at 800°C for 2h in different ratio of vapor atmosphere. As shown in Fig. 8, the H3-20 glass ceramic seal only exhibited the Al₂O₃ phase in air atmosphere, which is attributed to the strong stability of the H3 glass with 20wt% Al₂O₃ ceramic powders. However, in the vapor atmosphere, because of the 20wt% Al₂O₃ addition to H3 glass, a small reflection of wollastonite (CaSiO₃) phase appears after in 50vol% ratio of vapor, which is detected between $2\theta = 23^\circ - 25^\circ$ and $31^\circ - 33^\circ$ (PDF#72-2284) with unit cell dimensions of $a = 7.880\text{\AA}$, $b = 7.270\text{\AA}$, and $c = 7.070\text{\AA}$. In addition, the CaSiO₃ crystallization phases are effectively increased as the ratio of vapor increased to 90vol%, strongly suggesting the effect of vapor on H3-20 glass ceramic stabilization. This is due to the reaction of Ca²⁺ from the H3 glass with vapor and then generates Ca(OH)₂. Even worse, due to the exothermic reaction between Ca²⁺ and vapor will push H₂O molecules to penetrate loosely bound Ca²⁺ ions, breaking the Si-O bonds in the glass network and ultimately leading to hydroxylation of silicate glass [29]. Eventually, the Ca²⁺ ions rearrange with the hydroxylated silicate network to form CaSiO₃. The

reaction can be represented as (2), (3):



[Download: Download high-res image \(537KB\)](#)

[Download: Download full-size image](#)

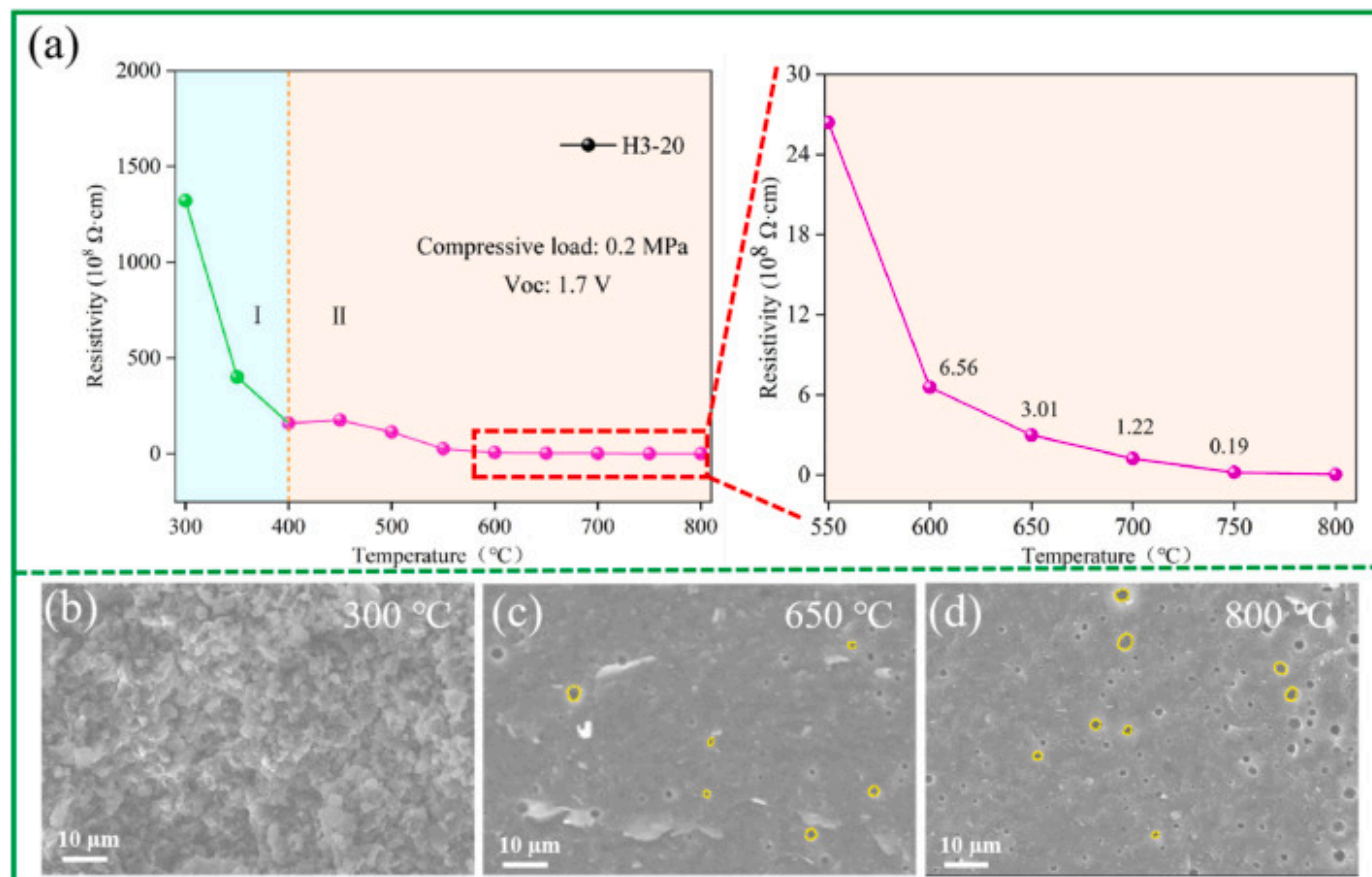
Fig. 8. XRD patterns of H3-20 glass ceramic seal after heat treated at 800°C for 2h various ratio of vapor atmosphere.

High hardness and melting point CaSiO₃ [30,31] will block the pores and leakage channels of the H3-20 glass ceramic seal and improve the gas tightness of the sealing component. Furthermore, the CTE of CaSiO₃ is about $9.4 \times 10^{-6} \text{K}^{-1}$ [32], which is close to that of SUS430 stainless steel interconnect ($10.4 \times 10^{-6} \text{K}^{-1}$) and NiO-YSZ cell anode ($12.4 \times 10^{-6} \text{K}^{-1}$) [33,34] that suggesting CaSiO₃ crystallization phases can further improve the thermal compatibility between the seals and other components of SOEC stack and enhance the thermal stability of the H3-20 glass ceramic seal.

3.2. Insulating performance of seals in 70vol% vapor atmosphere

Glass ceramic seals should have excellent H₂O corrosion resistance and electrical insulation in the vapor atmosphere to improve gas tightness in core and side cover for SOEC. Fig. 9 shows the resistivity and microstructure of H3-20 glass ceramic seal among the temperature range of 300–800°C in 70vol% H₂O vapor atmosphere under a constant voltage of 1.7V. As shown in Fig. 9(a), the resistivity of the H3-20 glass ceramic seal can be divided into two stages: temperature stage I that range of 300–400°C and temperature stage II that range of 400–800°C. The change in resistivity shows a trend of rapid decrease followed by stabilization. In temperature stage I, the insulation resistivity from $13 \times 10^{10} \Omega\text{cm}$ rapidly decreases to $1.6 \times 10^{10} \Omega\text{cm}$ while the temperature increases from 300°C to 400°C, However, when the H3-20 glass ceramic seal is heated higher than 600°C, the resistivity gradually decreases to $6.6 \times 10^8 \Omega\text{cm}$ and then tend to stable, which is about $0.2 \times 10^8 \Omega\text{cm}$ that as shown in the temperature stage II. Fortunately, the resistivity of the seals is higher than the minimum threshold of $10^4 \Omega\text{cm}$ required for SOEC operation [35]. There are four reasons for this phenomenon: firstly, it is due to weaken molecular motion and strong bond energies among Si-O covalent bonds lead to weak molecular motion in the H3 glass system at low temperature stage [36]. Besides, because of the presence of metal cations such as Ca²⁺ and Al³⁺ in the form of ionic bonds in the H3 glass network, the polar covalent bonds and ionic bonds in the seals have good bonding strength at low temperatures [37]. Furthermore, the covalent and trivalent characteristic of B₂O₃ components result in weak ionic mobility and enhanced bond energy in H3 glass at the low temperature stage, thereby improving the electrical insulation resistance of H3 glass [38]. However, when H3-20 glass ceramic seal in the ratio of 70vol% vapor at high temperature stage, the accelerated thermal motion of B₂O₃ molecule accelerates the migration speed of metal ions, result in an increase in the conductivity and a decrease in the resistance of the H3-20 glass ceramic seal [39]. Meanwhile, B₂O₃ reacts with vapor to form HBO₂ and H₃BO₃ [40]. The reaction can be represented as (4), (5):





[Download: Download high-res image \(752KB\)](#)

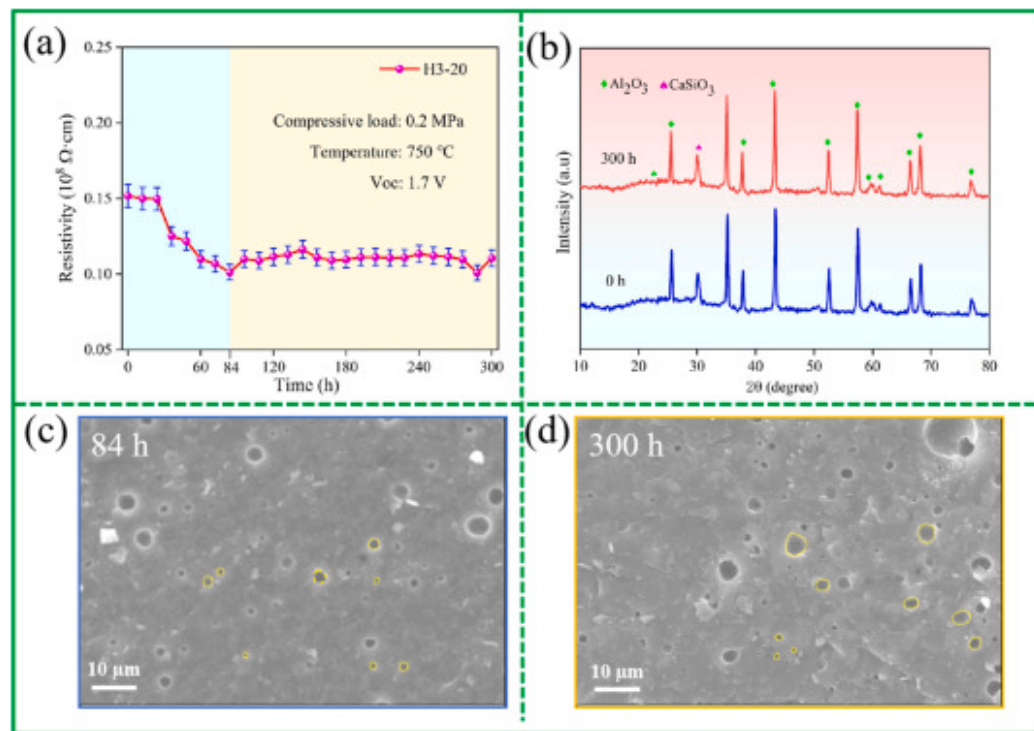
[Download: Download full-size image](#)

Fig. 9. Resistivity and SEM micrographs of H3-20 glass ceramic seal from 300 to 800°C under a constant voltage of 1.7V in 70vol% H₂O vapor atmosphere. (a) resistivity, (b–d) The SEM images of H3-20 seal at 300°C, 650°C and 800°C.

The erosion by vapor causes H⁺ ions to replace metal cations in H3 glass network, and the thermal motion of molecules in H3-20 seal intensifies leading to an increased ionization rate of metal ions that enhancing the H3-20 glass ceramic seal ion mobility [41,42]. From the SEM micrographs in Fig. 9(b–d), it is can be seen that as the temperature rises from 300°C to 650°C, the Al₂O₃ ceramic and H3 glass powders gradually evolve from a mixed stack state to a mutal molten and the H3-20 glass ceramic seal tends to dense without obvious leakage channels. However, it should be noted that circular pores appear on the surface of H3-20

glass ceramic seal at 650°C, which are caused by the volatilization of B₂O₃, reaction with vapor [43]. It can be seen from Fig. 9(d) that the number of pores corroded by vapor is significantly higher at 800°C than at 650°C. Fortunately, these closed pores on the surface of H3-20 glass ceramic seal don't connect together to form obvious small cracks, and thus, it still has good gas tightness and insulation in the wide temperature stage for SOEC.

Fig. 10 shows the resistivity, X-ray diffraction patterns and SEM micrographs of H3-20 glass ceramic seal at 750°C hold 300h under a voltage of 1.7V in the ratio of 70vol% vapor. It can be seen from Fig. 10(a) that the resistivity of H3-20 glass ceramic seal shows a trend of initially decreasing and then remaining stable with the extension of holding time. When the heat treat time extend to 84h, the resistivity decreases from $0.15 \times 10^8 \Omega\text{cm}$ to $0.1 \times 10^8 \Omega\text{cm}$, this is partly due to the free migration of metal ions such as Al³⁺ and Ca²⁺ in the H3-20 glass ceramic seal at 750°C which increases the conductivity of the seal. On the other hand, the Si-O-H unsaturated bonds on the glass surface can adsorb vapor and react to generate Si-OH hydrocarbon groups result in reducing the bridging oxygen bonds in the glass structure and lowering the electrical resistivity of the sealing material [44]. With the extension of hold time to 300h, the resistivity is stabilized at $0.1 \times 10^8 \Omega\text{cm}$. It implies that the network structure of internal fracture rearrangement in H3 glass will inhibit the further corrosion of vapor and promote the resistivity remains stability. Fig. 10(c, d) shows the microstructures of H3-20 glass ceramic seal hold on 84h and after 300h resistivity test. It can be seen that the pores of the H3-20 seal gradually increase, indicating that the corrosion process of vapor is consistent with the change in electrical resistivity.



[Download: Download high-res image \(609KB\)](#)

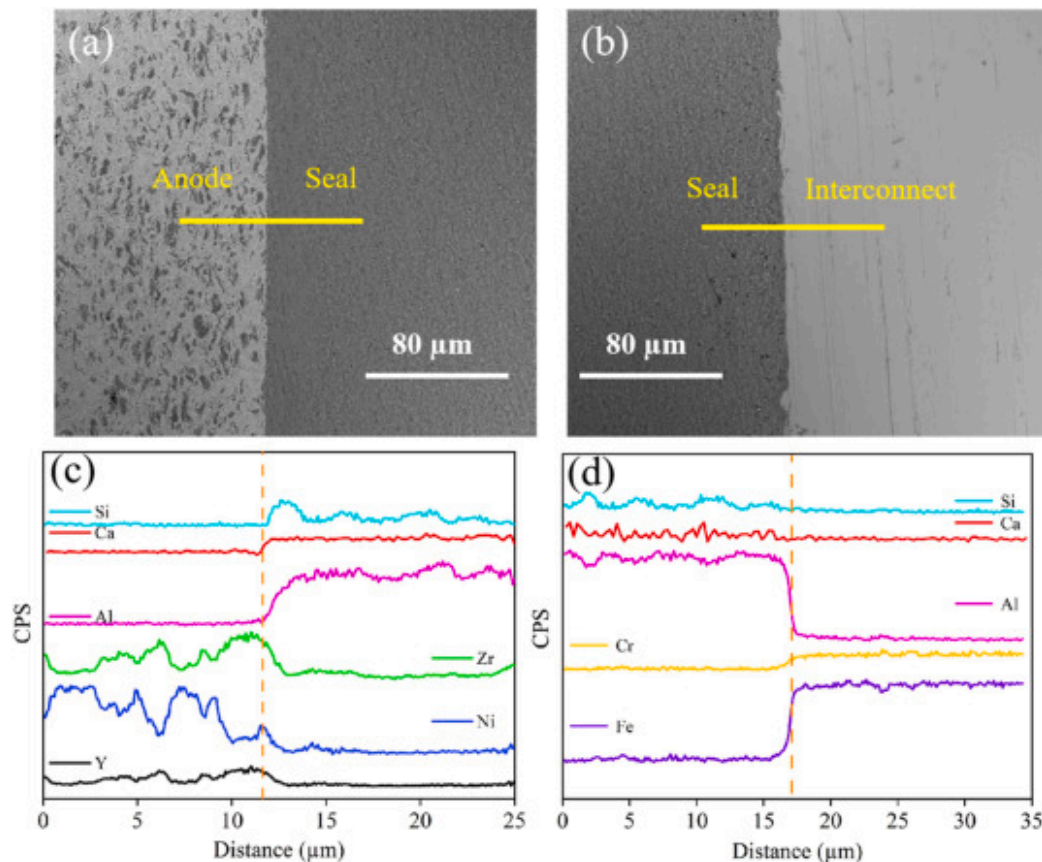
[Download: Download full-size image](#)

Fig. 10. Resistivity, XRD and SEM micrographs of H3-20 glass ceramic seal under a voltage of 1.7V at 750°C for 300h in the ratio of 70vol% vapor. (a) Resistivity, (b) XRD before and after test, (c) SEM for 84h and 300h.

Fig. 10(b) shows the XRD patterns of H3-20 glass ceramic seal while it hold on 0h and 300h test at 750°C. It is obvious from Fig. 10(b) that CaSiO_3 phase (PDF#72-2284) is clearly detected in H3-20 glass ceramic seal at 2θ is 28° – 32° , and the CaSiO_3 height of hold 300h is higher than 0h for H3-20 seal which implies long term holding at 750°C is more likely to promote the production of CaSiO_3 . There is no new phase appearance in these processes, which means that H3-20 glass ceramic seal has long term insulation stability. This is due to the mobility of Ca^{2+} approaching 1 will cause it to migrate quickly initially and then stably in the H3-20 glass ceramic seal, when the CaSiO_3 formed finally means the corrosion rate of vapor gradually decreases. The migration of Ca^{2+} and the rearrangement of Si-O bonds may also be the reasons for the initial decrease and subsequent stabilization of resistivity [45].

3.3. Interfacial and thermal cycling stability of the glass-ceramic seals

In order to investigate the bonding between H3-20 glass ceramic seal and SOEC adjacent joints to observe any chemical reaction is occurred or not, the interfacial morphology after 100h heat treated under 0.2MPa at 750°C in the 70vol% vapor is shown in [Fig. 11](#). It is observed from [Fig. 11](#)(a, b) that the H3-20 glass ceramic seal construct well adherence and crack-free interfaces, which can effectively lower the leakage rates and provide sufficient joining strength between adjacent components. To better understand the nature of the sealing interface, EDS line scan has been employed to analyze the composition of the interfacial area, as shown in [Fig. 11](#)(c and d). Ca, Si and Al elements belonging to glass were hardly detected in Fe-16Cr interconnect or cell anode. The corresponding EDS analysis indicated that the element diffusion between the adjacent H3-20 glass ceramic seal interface is difficult in relatively short heat treatment time. All in all, it is concluded that the element diffusion and chemical interaction may occur under higher operation temperature and longer time.



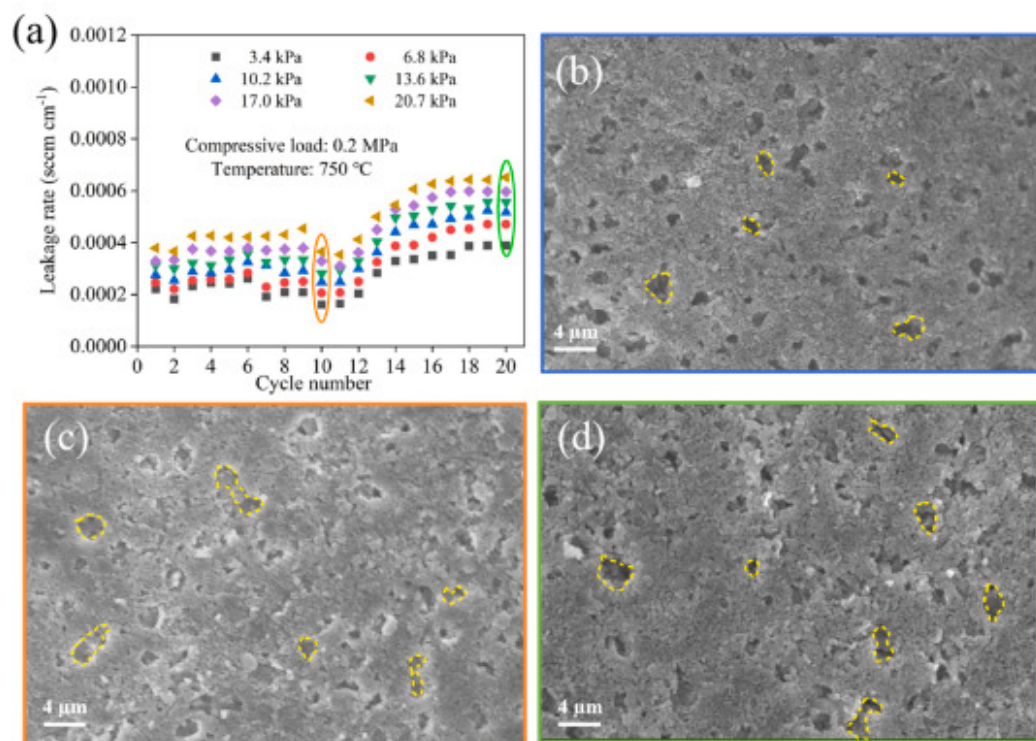
[Download: Download high-res image \(762KB\)](#)

[Download: Download full-size image](#)

Fig. 11. SEM micrographs and EDS compositional profiles across the interface between (a, c) H3-20 seal and cell anode, and (b, d) interconnect after exposure to 70vol% water vapor at 750°C for 100h.

Fig. 12 shows the leakage rate and micrographs of the H3-20 glass ceramic seal during the thermal cycle test under a compressive load of 0.2MPa and different input gas pressure in 70vol% vapor. As shown in Fig. 12(a), the leakage rate is stabilized at 0.0007 sccm cm⁻¹ during 20th thermal cycles under input gas pressure of 20.7kPa. However, it should be noted that the leakage rate decreased from 0.00048 sccm cm⁻¹ to 0.00035 sccm cm⁻¹ during the thermal cycle increased from 10th to 12th under input gas pressure of 20.7kPa. This may be due to the increasing content of vapor gradually formed a film that fill the gaps between the

seals and interconnect, reducing the leakage rate. As the thermal cycles increased from 16th to 20th, the leakage rate slightly increase and then stabilizes and the maximum leakage rate is approximately $0.00064 \text{ sccm cm}^{-1}$. It is attributed to the continuous frequent temperature change gradients will cause internal stress at the interface between the seals and stainless steel, resulting in leakage channels and gas tightness. Fig. 12(b–d) shows the micrographs of H3-20 glass ceramic seal 0th, 10th and 20th thermal cycle leakage rate test. It can be seen that the pores are uniformly distributed on the H3-20 seal surface initially. With the number of thermal cycles increase to 10th, because the extension of corrosion by vapor, the pores present a disordered arrangement and the H3-20 glass ceramic seal surface became rough, the pore size remained unchanged at approximately $1.3 \mu\text{m}$. When the H3-20 glass ceramic seal undergo 20th thermal cycles, because of the continuous corrosion of vapor enlarged these pores size that is approximately $2.6 \mu\text{m}$ and these pores tend to connect each other [46]. Based on the results of the leakage rate test, it can be concluded that the leakage rate gradually stabilized while the thermal cycle increased to 17th, indicating that the corrosion effect of vapor on the seals ceased.



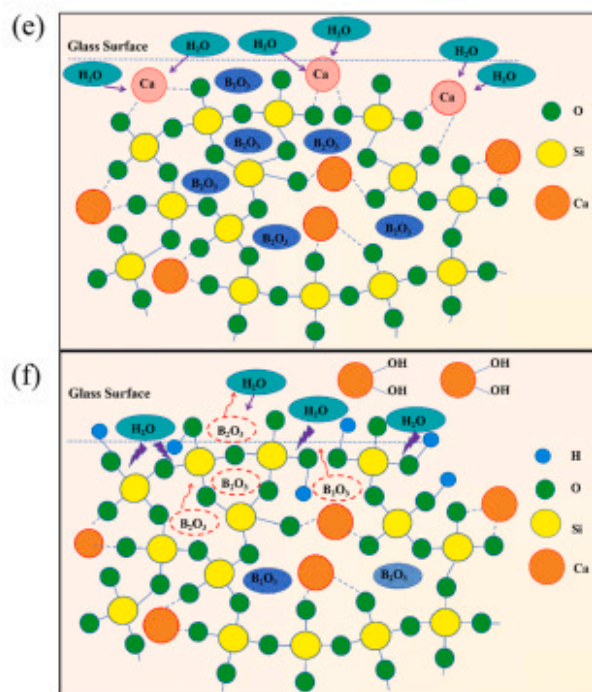
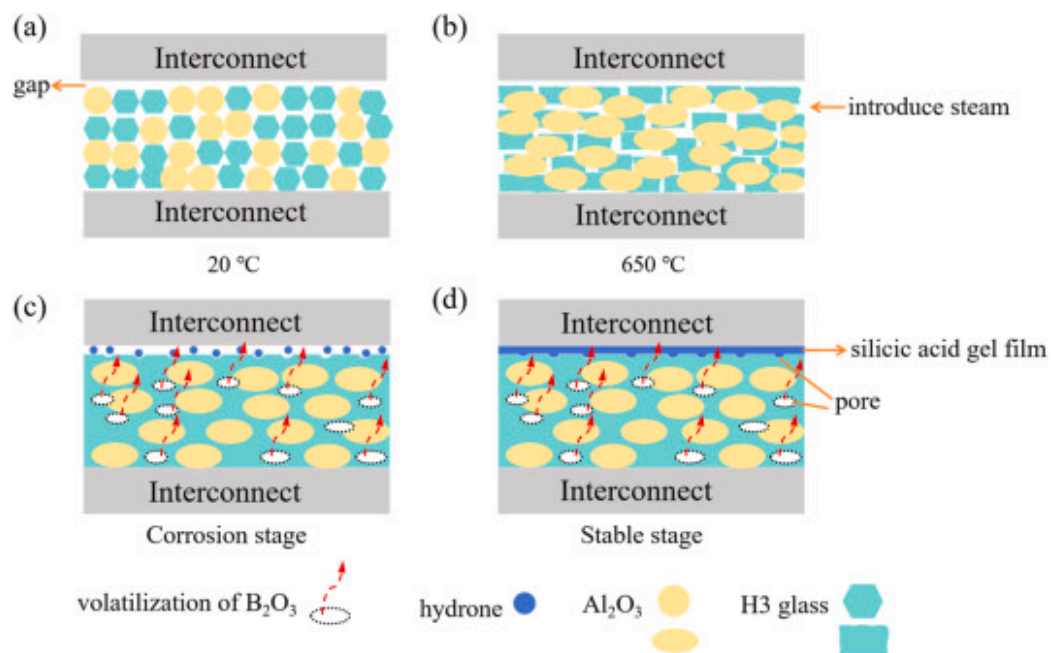
[Download: Download high-res image \(876KB\)](#)

[Download: Download full-size image](#)

Fig. 12. Leakage rates of H3-20 seal over 20 thermal cycles under a compressive load of 0.2 MPa, 70vol% steam ratio and input gas pressures from 3.4 kPa to 20.7 kPa at 750 °C (a), SEM micrographs of H3-20 seals after 0 thermal cycle (b), 10 thermal cycles (c), 20 thermal cycles (d).

3.4. Sealing mechanism of H3-20 glass ceramic seal

The reason for excellent sealing performance of H3-20 glass ceramic seal is the significant anti corrosion effect on vapor. Based on the investigation of various properties of the H3-20 seals, the vapor corrosion process can be divided into three stages: initial stage, corrosion stage and stable stage. Fig. 13 shows the schematic diagram of the vapor corrosion process for H3-20 glass ceramic seal. As shown in Fig. 13(a and b), when the temperature gradually increases from room temperature to 650 °C, H3-20 glass ceramic seal begins to soften lead to water vapor introduced at these gaps between interconnect and H3-20 seal form corrosion point. When the temperature rises from 650 °C to 800 °C the internal bonding of the seal becomes tighter because of the fluidity of the H3 glass, this is consistent with the leakage rate test results at the range of 650–800 °C. Besides, at high temperature stage (corrosion and stable stage), many surface pores are formed and become more susceptible due to the volatilization of organic binders and B₂O₃ to concentrated corrosion by vapor, as shown in Fig. 13(c). According to relevant literature, vapor can corrode the glass surface by attacking the glass network structure, making the corroded areas more prone to new wollastonite (CaSiO₃) crystallization [47]. When the temperature increases, the intensified molecular thermal motion leads to more frequent and energetic collisions between molecules, accelerating the formation rate of the silicic acid gel film and increasing the concentration of H₂O molecules in contact with the glass surface provides more reactants. The formation of the silicic acid gel film also fills the gaps between the seal and the adjacent components, as well as the pores on the glass surface, thereby improving the sealing effect, as shown in Fig. 13(d).



Download: [Download high-res image \(893KB\)](#)

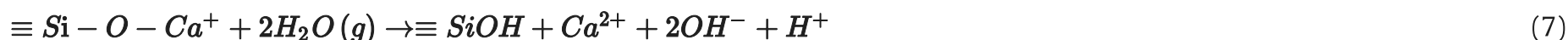
[Download: Download full-size image](#)

Fig. 13. Schematic diagram illustration of the steam corrosion process of H3-20 seal as the temperature rises from 20°C to 800°C. (a, b) Initial stage, (c) corrosion stage, (d) stable stage, (e, f) corrosion mechanism diagrams.

Fig. 13(e, f) shows the H3 glass internal structure corrosion mechanism of glass-based seals, calcium ions on the glass surface are attacked by H₂O molecules and undergo ion exchange with hydrogen ions [[48], [49], [50]]. The original positions of calcium ions are replaced by hydrogen ions, resulting in weakened surface strength. With continuous H₂O molecule ingress, the glass becomes more susceptible to corrosion. This process can be represented as (4), (5):



Since the H3 glass system not only possesses a continuous network structure but also contains a type of broken terminal structure, this terminal structure reacts with vapor [51].



Ultimately, a large number of H₂O molecules directly disrupt the Si-O bonds in the glass, leading to corrosion:



Calcium hydroxide is a basic oxide, which reacts with the acidic oxide silicon dioxide at high temperatures to form the neutralization product CaSiO₃. This reaction explains the formation of CaSiO₃ observed in the tests. However, this compound has not been found to have any adverse effects on the seals and is present in relatively low amounts.

4. Conclusions

SiO₂-CaO-B₂O₃-Al₂O₃ glass ceramic seal for SOECs have been prepared and the sealing property, thermal cycling stability, interfacial chemical compatibility, and electrical insulation of the seal as a atmosphere of vapor ratios have been systematically studied. Some conclusions can be indicated from this study, which are presented as follows:

1) H3-20 glass ceramic seal exhibits excellent sealing performance during the temperature stage of 650–800°C.

Because the CaSiO₃ silicic acid gel film produced in the seal while the vapor ratios ranging from 50vol% to

90vol%, the leakage rate of H3-20 glass ceramic seal is only 0.00043 sccm cm⁻¹ under a input pressure of 20.7kPa.

- 2) The insulation resistivity of H3-20 glass ceramic seal is about $2 \times 10^7 \Omega \text{cm}$ in the temperature range of 300–800°C under 70vol% H₂O vapor atmosphere and a constant voltage of 1.7V. With the extension to 300h, the resistivity of H3-20 glass ceramic seal is stabilized at $1 \times 10^7 \Omega \text{cm}$, which is attributed to the migration of Ca²⁺ and the rearrangement of Si-O bonds preventing further corrosion by vapor.
- 3) Interfacial microstructure and chemical analysis are conducted to investigate the sealing performance stability of the seals, showing well bonding interface in the interconnect/seal/cell. Furthermore, leakage rates below 0.0008 sccm cm⁻¹ is obtained after 20 thermal cycles under input gas pressure between 3.4kPa and 20.7kPa.
- 4) Calcium ions on the glass surface are attacked by H₂O molecules that lead to CaSiO₃ crystallization and silicic acid gel are film produced in H3 glass ceramic seal. These substances can fill the gaps between the seal and the adjacent components as well as the pores on the glass surface, thereby improving the sealing performance.

CRedit authorship contribution statement

Xiaochun Wang: Conceptualization. **Yaocheng Wang:** Writing – original draft. **Shen Jiang:** Formal analysis. **Siyuan Dang:** Formal analysis. **Li Li:** Writing – review & editing. **Jian Pu:** Data curation. **Jiangtao Feng:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by National Science Foundation of China ([52202094](#)), National Key Research & Development Project of China ([2022YFB4002203](#)). Key Research & Development Project of Hubei Province ([2023BAB115](#)). Jiangsu Province industrial foresight and key core technology research-key project, grant number [BE2023092-1](#). Changzhou Applied Research Project ([CJ20241075](#)). Zhongwu Talent of Jiangsu University of technology. The SEM and XRD characterizations assisted by the Analytical and Testing Center of Jiangsu University of Technology. The SOFC stack test platform assisted by Jiangsu Jingci intelligent sensing Technology Research Institute Co. , LTD.

[Recommended articles](#)

References

- [1] M. Reiser, A. Aphale, P. Singh
Solid oxide electrochemical systems: material degradation processes and novel mitigation approaches J
Materials, 11 (11) (2018), p. 2169, [10.3390/ma11112169](#) ↗
2169
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [2] B. Saheli, K. Gurpreet, P. Gary, *et al.*
A critical review on cathode materials for steam electrolysis in solid oxide electrolysis J
Int. J. Hydrogen Energy, 48 (34) (2023), pp. 12541-12570, [10.1016/j.ijhydene.2022.11.307](#) ↗
[Google Scholar](#) ↗
- [3] A. Hauch, S.D. Ebbesen, S.H. Jensen, *et al.*
Solid oxide electrolysis cells: microstructure and degradation of the Ni/yttria-stabilized zirconia electrode J
J. Electrochem. Soc., 155 (11) (2008), Article B1184, [10.1149/1.2967331](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [4] L. Liu, W. Zheng, Z. Ma, *et al.*
Study on water corrosion behavior of ZrSiO₄ materials J
J. Adv. Ceram., 7 (2018), pp. 336-342, [10.1007/s40145-018-0283-3](#) ↗

[Google Scholar ↗](#)

- [5] T. Jin, K. Lu
Thermal stability of a new solid oxide fuel/electrolyzer cell seal glass J
J. Power Sources, 195 (1) (2009), pp. 195-203, [10.1016/j.jpowsour.2009.07.023 ↗](#)
[Google Scholar ↗](#)
- [6] Z. Qi, Z. Xueqin, M. Zhuang, *et al.*
Water vapor and CMAS corrosion tests of Y₂SiO₅/Si thermal and environmental barrier coating J
Heliyon (2022), Article 1, [10.21203/rs.3.rs-889492/v1 ↗](#)
[Google Scholar ↗](#)
- [7] M.K. Mahapatra, K. Lu, R.J. Bodnar
Network structure and thermal property of a novel high temperature seal glass J
Appl. Phys. A, 95 (2009), pp. 493-500, [10.1007/s00339-008-4926-z ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [8] M.K. Mahapatra, K. Lu, Jr W.T. Reynolds
Thermophysical properties and devitrification of SrO-La₂O₃-Al₂O₃-B₂O₃-SiO₂-based glass sealant for solid oxide fuel/electrolyzer cells J
J. Power Sources, 179 (1) (2008), pp. 106-112, [10.1016/j.jpowsour.2007.12.101 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [9] K. Lu, M.K. Mahapatra
Network structure and thermal stability study of high temperature seal glass
Jpn. J. Appl. Phys., 104 (7) (2008), [10.1063/1.2979323 ↗](#)
[Google Scholar ↗](#)
- [10] P. Błaszczak, M. Zajęc, A. Ducka, *et al.*
High-temperature Co-electrolysis of CO₂/H₂O and direct methanation over Co-impregnated SOEC. Bimetallic synergy between Co and Ni, J

Int. J. Hydrogen Energy, 47 (82) (2022), pp. 35017-35037, [10.1016/j.ijhydene.2022.08.057](https://doi.org/10.1016/j.ijhydene.2022.08.057) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [11] X. Wang, D.R. Ou, Z. Zhao, *et al.*
Stability of SrO-La₂O₃-Al₂O₃-SiO₂ glass sealants in high-temperature air and steam J

Ceram. Int., 42 (6) (2016), pp. 7514-7523, [10.1016/j.ceramint.2016.01.158](https://doi.org/10.1016/j.ceramint.2016.01.158) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [12] A.G. Sabato, A. Rost, J. Schilm, *et al.*
Effect of electric load and dual atmosphere on the properties of an alkali containing diopside-based glass sealant for solid oxide cells J

J. Power Sources, 415 (2019), pp. 15-24, [10.1016/j.jpowsour.2019.01.051](https://doi.org/10.1016/j.jpowsour.2019.01.051) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [13] R. Li, L. Peng, X. Wang, *et al.*
Investigating the performance of Glass/Al₂O₃ composite seals in planar solid oxide fuel cells J

Compos. B Eng., 192 (2020), Article 107984, [10.1016/j.compositesb.2020.107984](https://doi.org/10.1016/j.compositesb.2020.107984) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [14] S. Fu, F. Cheng, R. Li, *et al.*
Performance of Al₂O₃ reinforced BaO-SiO₂-CaO glass as sealing materials for solid oxide fuel cells J

Ceram. Int., 51 (2025), pp. 34106-34114, [10.1016/j.ceramint.2025.05.137](https://doi.org/10.1016/j.ceramint.2025.05.137) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [15] R. Li, Y. Lu, Y. Yu, *et al.*
Thermal performance of glass-based seals for reversible solid oxide cell: effect of Al₂O₃ ceramic addition on the crystallization behavior J

Ceram. Int., 50 (2024), pp. 6481-6489, [10.1016/j.ceramint.2023.11.393](https://doi.org/10.1016/j.ceramint.2023.11.393) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [16] R. Li, M. Tao, P. Wang, *et al.*

Effect of interconnect pre-oxidation on high-temperature wettability and mechanical properties of glass seals in SOFC J

J. Am. Ceram. Soc., 104 (12) (2021), pp. 6172-6182, [10.1111/jace.18024](https://doi.org/10.1111/jace.18024) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [17] Wang X. C., Wang Y. C., Xu S. P., et al. Gas-Tightness Testing Device for Seals: China, CN202420230737.1P. 2024-10-11.

[Google Scholar ↗](#)

- [18] Z.L. Yin, B.A. Liu, G.P. Feng

Derivation of a new expression for Kirchhoff's law based on field theory

Modern Electronics Technique, 34 (19) (2011), pp. 190-192, [10.16652/j.issn.1004-373x.2011.19.013](https://doi.org/10.16652/j.issn.1004-373x.2011.19.013) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [19] R. Wang, L. Ma, f Li, et al.

Preparation and optimization of solid oxide electrolysis cells J

Clean Coal Technology, 30 (S1) (2024), pp. 99-103, [10.13226/j.issn.1006-6772.24040807](https://doi.org/10.13226/j.issn.1006-6772.24040807) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [20] R. Li, J. Yang, D. Yan, et al.

Performance of Al₂O₃ particle reinforced glass-based seals in planar solid oxide fuel cells J

Ceram. Int., 47 (6) (2021), pp. 7504-7510, [10.1016/j.ceramint.2020.10.256](https://doi.org/10.1016/j.ceramint.2020.10.256) ↗



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [21] X. Wang, Z. Kou, H. Wu, et al.

Improvement of thermal cycle stability of YSZ-glass composite seals for intermediate temperature solid oxide fuel cell J

Ceram. Int., 49 (22) (2023), pp. 36734-36742, [10.1016/j.ceramint.2023.08.359](https://doi.org/10.1016/j.ceramint.2023.08.359) ↗



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [22] D. Rebiscoul, P. Frugier, S. Gin, et al.

Protective properties and dissolution ability of the gel formed during nuclear glass alteration J

J. Nucl. Mater., 342 (1–3) (2005), pp. 26–34, [10.1016/j.jnucmat.2005.03.018](https://doi.org/10.1016/j.jnucmat.2005.03.018) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [23] D. Rebiscoul, A. Van der Lee, F. Rieutord, *et al.*
Morphological evolution of alteration layers formed during nuclear glass alteration: new evidence of a gel as a diffusive barrier J

J. Nucl. Mater., 326 (1) (2004), pp. 9–18, [10.1016/j.jnucmat.2003.10.015](https://doi.org/10.1016/j.jnucmat.2003.10.015) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [24] S. Gin, P. Jollivet, M. Fournier, *et al.*
Origin and consequences of silicate glass passivation by surface layers J

Nat. Commun., 6 (1) (2015), p. 6360, [10.1038/ncomms7360](https://doi.org/10.1038/ncomms7360) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [25] Y. Zhao, Z. Shen, Z. Tian, *et al.*
Corrosion behaviour of low dielectric glass fibres in hydrochloric acid J

J. Non-Cryst. Solids, 511 (2019), pp. 212–218, [10.1016/j.jnoncrysol.2018.12.039](https://doi.org/10.1016/j.jnoncrysol.2018.12.039) ↗



[View PDF](#) [View article](#) [Google Scholar](#) ↗

- [26] Y. Nie
Study on the High-Temperature Corrosion Mechanism of Glass D

Fuzhou University (2018)

[Google Scholar](#) ↗

- [27] R. Li, X. Wang, L. Peng, *et al.*
Thermal cycling stability of novel hexagonal boron nitride (h-BN)/glass compressive seals for planar intermediate temperature solid oxide fuel cells J

J. Alloys Compd., 843 (2020), Article 155620, [10.1016/j.jallcom.2020.155620](https://doi.org/10.1016/j.jallcom.2020.155620) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [28] Q.T. Duan

Study on the Reaction of Water Vapor with Glass Melt Under Full-Oxygen Combustion Conditions and Its Effects on Glass Structure and Properties D

Hainan University (2015)

[Google Scholar ↗](#)

[29] E.M. Pierce, P. Frugier, L.J. Criscenti, *et al.*

Modeling interfacial glass-water reactions: recent advances and current limitations J

Int. J. Appl. Glass Sci., 5 (4) (2014), pp. 421-435, [10.1111/ijag.12077 ↗](#)

[Google Scholar ↗](#)

[30] Hoc N. Quang, N. Duc Trung, N. Ngoc Thang, *et al.*

Melting curves of MgSiO₃ and CaSiO₃ perovskites with cubic structure at extreme conditions

Jpn. J. Appl. Phys., 136 (4) (2024), [10.1063/5.0207166 ↗](#)

[Google Scholar ↗](#)

[31] J. Zhao, N. Chen, H. Xiao, *et al.*

Fabrication of CaSiO₃ LTCC high aspect ratio microstructure by laser chemical assisted micro-milling J

Ceram. Int., 49 (23) (2023), pp. 38139-38147, [10.1016/j.ceramint.2023.09.144 ↗](#)



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[32] M.K. Mahapatra, K. Lu

Glass-based seals for solid oxide fuel and electrolyzer cells—a review J

Mater. Sci. Eng. R Rep., 67 (5–6) (2010), pp. 65-85, [10.1016/j.mser.2009.12.002 ↗](#)



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[33] Q. Zhang, H. Zhuang

(third ed.), Brazing Manual, 49, M. Welding Technology (2020), p. 102, [10.13846/j.cnki.cn12-1070/tg.2020.02.029 ↗](#)
02

[View in Scopus ↗](#) [Google Scholar ↗](#)

[34] X. Wang

Structural Design and Performance Study of YSZ-Glass Composite Sealing Materials for intermediate-temperature Solid Oxide Fuel Cells D

Huazhong University of Science and Technology (2019)

[Google Scholar ↗](#)

- [35] C. Lara, M.J. Pascual, R. Keding, *et al.*
Electrical behaviour of glass-ceramics in the systems RO-BaO-SiO₂ (R= Mg, Zn) for sealing SOFCs J
J. Power Sources, 157 (1) (2006), pp. 377-384, [10.1016/j.jpowsour.2005.07.084 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [36] S. Nagai, K. Nakao, Y. Hayashi, *et al.*
Low-temperature bonding of glasses: the role of plasma-engrafted surface amino groups J
J. Am. Ceram. Soc., 107 (5) (2024), pp. 2877-2887, [10.1111/jace.19684 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [37] X. Wang, X. Zhang, Y. Sun, *et al.*
Resistivity modulation of perovskite samarium nickelate with high-valence cations and the underlying mechanism J
Appl. Surf. Sci., 624 (2023), Article 157103, [10.1016/j.apsusc.2023.157103 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [38] S.Y. Gao, X.N. Liu, D.N. Gao, *et al.*
Development of electrically insulating sealing glasses in the Li₂O-B₂O₃-SiO₂ system
J. Shaanxi Univ. Sci. Technol. Nat. Sci. Ed. (5) (2005), pp. 50-52
[Google Scholar ↗](#)
- [39] X. Xi, Z. Wang, *et al.*
Effect of B₂O₃ on the structure and electrical properties of high-temperature sealing glass J
J. Ceram. Sci. Technol. (2023), [10.4416/JCST2023-00005 ↗](#)
[Google Scholar ↗](#)

- [40] Q. Shan, J. Hu, H. Li, *et al.*
Research progress on self-healing silicon carbide ceramic matrix composites for aeroengines J
Aeronautical Manufacturing Technology, 66 (19) (2023), pp. 48-58, [10.16080/j.issn1671-833x.2023.19.048](https://doi.org/10.16080/j.issn1671-833x.2023.19.048) ↗
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [41] Z. Zhang, Q. Wang, F. Wang, *et al.*
Effect of aqueous environment on wear resistance of dental glass–ceramics J
BMC Oral Health, 22 (1) (2022), p. 143, [10.1186/s12903-022-02183-5](https://doi.org/10.1186/s12903-022-02183-5) ↗
 [View PDF](#) [View article](#) [Google Scholar ↗](#)
- [42] J.F. Ortiz-Mosquera, A.M. Nieto-Munoz, A.C.M. Rodrigues
Influence of Al³⁺ on glass-forming ability, structural and electrical properties of the Na_{3.4}Sc₂Si_{0.4}P_{2.6}O₁₂ superionic conductor J
J. Alloys Compd., 850 (2021), Article 156670, [10.1016/j.jallcom.2020.156670](https://doi.org/10.1016/j.jallcom.2020.156670) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [43] Z.Y. Wei, S.X. Gu, X.W. Wang, *et al.*
Interaction process and reaction mechanisms between glass and water J
Bulletin of the Chinese Ceramic Society, 41 (11) (2022), pp. 4049-4055, [10.16552/j.cnki.issn1001-1625.2022.11.030](https://doi.org/10.16552/j.cnki.issn1001-1625.2022.11.030) ↗
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [44] D. Wu, H. Ren, X. Huang, *et al.*
Strengthening functional properties and competitive crystallization behavior of La₂O₃-BaO-CaO-Al₂O₃-B₂O₃-SiO₂ glass-ceramic for solid oxide fuel cells: agglomeration effect of rare-earth oxide J
J. Non-Cryst. Solids, 632 (2024), Article 122933, [10.1016/j.jnoncrysol.2024.122933](https://doi.org/10.1016/j.jnoncrysol.2024.122933) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [45] G. Zhang, Q. Xue, L. Li, *et al.*
Electrical conductivity and ion diffusion coefficient of CaO-Al₂O₃-SiO₂ melts J
J. Univ. Sci. Technol. Beijing, 34 (11) (2012), pp. 1250-1255, [10.13374/j.issn1001-053x.2012.11.011](https://doi.org/10.13374/j.issn1001-053x.2012.11.011) ↗
[View in Scopus ↗](#) [Google Scholar ↗](#)

- [46] W. Zhou, Z. Niu, X. Chen, *et al.*
Synergistic effect of water vapour on the thermal corrosion of CFAS melt to Yb₂Si₂O₇ environmental barrier coating material J
Corros. Sci., 225 (2023), Article 111625, [10.1016/j.corsci.2023.111625](https://doi.org/10.1016/j.corsci.2023.111625) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [47] R. Müller, E.D. Zanotto, V.M. Fokin
Surface crystallization of silicate glasses: nucleation sites and kinetics J
J. Non-Cryst. Solids, 274 (1–3) (2000), pp. 208–231, [10.1016/S0022-3093\(00\)00214-3](https://doi.org/10.1016/S0022-3093(00)00214-3) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [48] R.H. Do Remus, Y. Mehrotra, W.A. Lanford, *et al.*
Reaction of water with glass: influence of a transformed surface layer J
J. Mater. Sci., 18 (1983), pp. 612–622, [10.1007/BF00560651](https://doi.org/10.1007/BF00560651) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [49] S. Fujita, A. Sakamoto, M. Tomozawa
Behavior of water in glass during crystallization J
J. Non-Cryst. Solids, 320 (1–3) (2003), pp. 56–63, [10.1016/j.physb.2010.03.044](https://doi.org/10.1016/j.physb.2010.03.044) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [50] F.H. El-Batal, E.M.A. Khalil, Y.M. Hamdy, *et al.*
Infrared reflection spectroscopy for precise tracking of corrosion behavior in 3d-transition metals doped binary lead silicate glass
J. Physica B: Condensed Matter, 405 (12) (2010), pp. 2648–2653, [10.1016/s0022-3093\(03\)00077-2](https://doi.org/10.1016/s0022-3093(03)00077-2) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [51] C.Y. Wang, Y. Tao
Handbook of Glass Properties and Processing
Chemical Industry Press, Beijing (2014)
[Google Scholar](#) ↗

Cited by (0)

[View Abstract](#)

© 2025 Elsevier Ltd and Techna Group S.r.l. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.